$$P(a) = \sum_b \psi^*(a,b)\psi(a,b).$$

This is just a diagonal entry in the density matrix:

$$P(a) = \rho_{aa}. \tag{7.22}$$

Here are some properties of density matrices:

- Density matrices are Hermitian:

$$\rho_{aa'} = \rho^*_{a'a}.$$

- The trace of a density matrix is 1:

$$Tr(\rho) = 1.$$

  Eq. 7.22 should help make this clear because the left side is a probability.

- The eigenvalues of the density matrix are all positive and lie between 0 and 1. It follows that if any eigenvalue is 1, all the others are 0. Can you interpret this result?

- For a pure state:

$$\rho^2 = \rho$$

$$Tr(\rho^2) = 1$$